

HW 3 Math 638 Differential Geometry II, Winter 2026

Due on Wednesday, January 28, 2026 on class.

Q3.1 ([Lee, 2nd, 3-4, p.75]) Show that TS^1 is diffeomorphic to $S^1 \times \mathbb{R}$.

Q3.2 ([Lee, 2nd, 8-2, p.199]) (Euler's homogeneous function theorem) Let c be a real number, and let $f : \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}$ be a smooth function that is positively homogeneous of degree c , $f(\lambda x) = \lambda^c f(x)$ for all $\lambda > 0$ and $x \in \mathbb{R}^n \setminus \{0\}$. Prove that $Vf = cf$, where V is the Euler vector field defined in [Lee, 2nd, Example 8.3].

Q3.3 ([Lee, 2nd, 8-10, p.201]) Let $M^2 = \{(x, y) \in \mathbb{R}^2, x > 0, y > 0\}$ be the open submanifold, and let $F : M \rightarrow M$ be the map $F(x, y) = (xy, y/x)$. Show that F is a diffeomorphism, and compute F_*X and F_*Y , where

$$X = x\partial_x + y\partial_y, \quad Y = y\partial_x.$$

Q3.4 ([Lee, 2nd, 8-12, p.201]) Let $F : \mathbb{R}^2 \rightarrow \mathbb{RP}^2$ be the smooth map $F(x, y) = [x, y, 1]$, and let $X \in \mathfrak{X}(\mathbb{R}^2)$ be defined by $X = x\partial_y - y\partial_x$. Prove that there is a vector field $Y \in \mathfrak{X}(\mathbb{RP}^2)$ that is F -related to X , and compute its coordinate representation in terms of each of the charts defined in [Lee, 2nd, Example 1.5].

Q3.5 ([Lee, 2nd, 8-16(a)(b), p.201]) For each of the following pairs of vector fields X, Y defined on $\mathbb{R}^3$, compute the Lie bracket $[X, Y]$.

- (a) $X = y\partial_z - 2xy^2\partial_y, \quad Y = \partial_y;$
(b) $X = x\partial_y - y\partial_x, \quad Y = y\partial_z - z\partial_y.$

Q3.6 ([Lee, 2nd, 8-20, p.202]) Let $A \subset \mathfrak{X}(\mathbb{R}^3)$ be the subspace with basis $\{X, Y, Z\}$ where

$$X = y\partial_z - z\partial_y, \quad Y = z\partial_x - x\partial_z, \quad Z = x\partial_y - y\partial_x.$$

Show

- (6a) A is a Lie subalgebra of Lie algebra $\mathfrak{X}(\mathbb{R}^3)$ of vector fields.
(6b) Lie algebra A is isomorphic to the Lie algebra $\mathbb{R}^3$ with the cross product Lie bracket.

Q3.7 ([Lee, 2nd, 8-31, p.204]) Let $\mathfrak{g}$ be a Lie algebra. A linear subspace $\mathfrak{h} \subset \mathfrak{g}$ is called an ideal in $\mathfrak{g}$ if $[X, Y] \in \mathfrak{h}$ whenever $X \in \mathfrak{h}$ and $Y \in \mathfrak{g}$.

(7a) Show that if $\mathfrak{h}$ is an ideal in $\mathfrak{g}$, then the quotient space $\mathfrak{g}/\mathfrak{h}$ has a unique Lie algebra structure such that the projection $\pi : \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{h}$ is a Lie algebra homomorphism.

(7b) Show that a subspace $\mathfrak{h} \subset \mathfrak{g}$ is an ideal if and only if it is the kernel of a Lie algebra homomorphism.